

Pratyush Agarwal

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TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, JavaScript, SQL, Bash/Shell Scripting

Frontend: React.js, Next.js, HTML, CSS

Backend and APIs: Node.js, FastAPI, REST APIs, GraphQL, SQLAlchemy, Postman

Databases: PostgreSQL, MySQL, SQL Server, DynamoDB, Redis, BigQuery

Cloud and DevOps: AWS (EC2, S3, Lambda, API Gateway, IAM), Azure, GCP, Docker, Kubernetes, Git/GitHub, CI/CD, Microservices, Agile/Scrum

CS Foundations: Data Structures, Algorithms, Object-Oriented Programming, System Design

AI: LLM Applications (Claude API, AWS Bedrock, Cursor, Strands), Prompt Engineering, LangChain, LangGraph

PROJECTS

Mnemo: Personal Knowledge Engine | *TypeScript, NestJS, Python, FastAPI, PostgreSQL, Docker* [\[GitHub\]](#)

- Building a two-service "second brain" app: a TypeScript/NestJS API for notes, tasks, auth, and kanban, and a Python/FastAPI AI service communicating over an HTTP contract, orchestrated with Docker Compose.
- Architected with clean layering (thin routers, service logic, repository DB access) on Postgres and pgvector, evolving toward semantic search, GraphRAG over Neo4j, and an MCP-exposed agent.

TickerSense | *React, Next.js, Node.js, Claude API, PostgreSQL* [\[GitHub\]](#)

- Built a full-stack LLM platform with React, Next.js, and Node.js unifying SEC filings, fundamentals, and disclosures into one interface; placed 2nd at the Claude Builder Club Hackathon (UIUC).

TripTok | *FastAPI, PySpark, AWS, Python* [\[GitHub\]](#)

- Built an end-to-end pipeline processing 25GB+ of NYC Taxi data with PySpark and AWS S3, serving ML models via a FastAPI MCP server with AWS Bedrock and Strands Agents.

EXPERIENCE

ADM Aug 2026 – Present
Data Engineer Intern Champaign, IL

- Supporting ADM's Process Development and Carbohydrates teams by automating data cleaning, analysis, and reporting workflows from lab experiments for Power BI/Excel Reporting, targeting ~15 hours/week in researcher time savings.

Kohler Co. [Digital Transformation Team] Jun 2025 – May 2026
Data Engineer Intern Champaign, IL

- Designed and implemented production ETL pipelines with Azure Data Factory and Databricks, ingesting SharePoint and on-prem SQL Server data into Parquet-based curated layers for downstream systems.
- Integrated on-prem operational databases with Azure cloud platforms and introduced Git-based version control to the team, improving pipeline governance across the org.

Apna [Trust and Safety Team] Oct 2022 – Aug 2024
Data Engineer Bangalore, India

- Built an API-driven user verification pipeline using Python, REST APIs, and Postman against Government of India identity services, halving processing time for ~1,200 users weekly.
- Optimized high-cost BigQuery workloads through partition-filtering and scope-limiting, cutting warehouse spend by ~\$2,500 monthly and improving query execution time by ~20%.

Google [Channel Sales Activation Team] Oct 2021 – Apr 2022
Strategy and Analytics Intern Singapore

- Engineered web scraping automation with Python, ChromeDriver, and Selenium, plus the database and supporting infrastructure to capture daily website metrics; extended the role from 1 to 7 months.

EDUCATION

University of Illinois at Urbana-Champaign Aug 2024 – May 2026
Master of Science in Information Management; GPA: 3.75/4.0 Champaign, IL

Manipal University Jaipur Jul 2018 – Jul 2022
Bachelor of Technology in Computer and Communication Engineering; GPA: 8.53/10.0 Jaipur, India